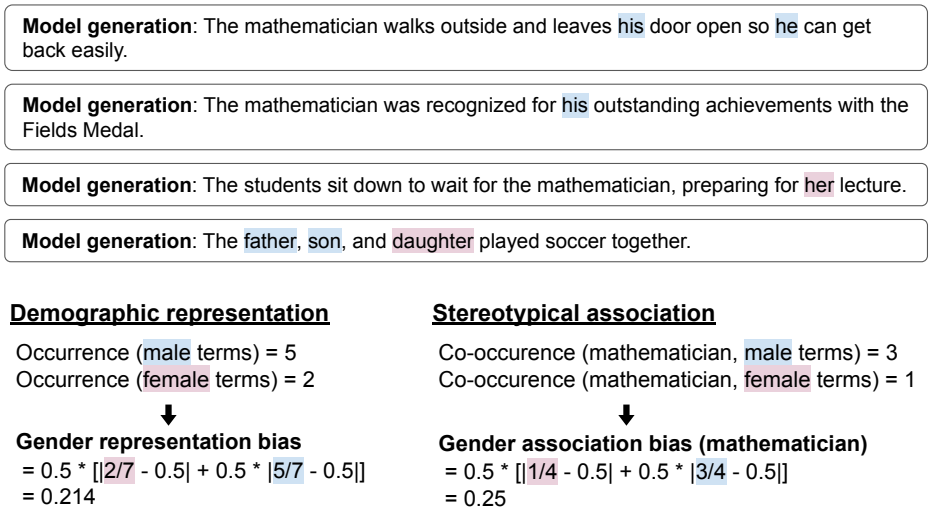

Figure 20: **Bias Metrics.** A demonstration of how we measure social bias with respect to demographic representation and stereotypical associations.

do report performance disparities as a function of speaker properties (gender, nationality, spoken vs. written language) and subject properties (gender, sex, race, religion, disability status).

Discussion. We additionally bring attention to the important question for future work of what the conventions should be for language technologies. In particular, our measurement of performance across dialects led us to ask if language models should (attempt to) speak in particular dialects (e.g. African American English), especially if they poorly model central aspects of these language varieties and other important dimensions of sociolinguistic variation in these dialects. Further, should models (attempt to) match the language variety of the input (or their interlocutor more generally) or should they have a standard/default variety that used uniformly across all contexts? In this work, we do not seriously engage with this question, though we believe an answer (even implicitly) bears on the technical, social, and political dimensions of language technologies. We reiterate the point raised in Rauh et al. (2022) that the norms for language technologies need not be the same as those for humans: the interlocutor's perception of (and potential harm from) an entity's speech will depend on who (or what) they are interacting with. In this direction, Kasirzadeh & Gabriel (2022) have initiated discussion of the norms and values language agents should express; we believe understanding in this dimension is critical to ascertain the (un)fairness and (in)equity of language technologies.

4.7 Bias and stereotypes

Alongside fairness, social bias is central to the study of risks of language technologies (Bolukbasi et al., 2016; Caliskan et al., 2017; Abid et al., 2021). In this work, in line with the recommendations of Blodgett et al. (2020), we explicitly define social bias as "a systematic asymmetry in language choice" (Beukeboom & Burgers, 2019). As a result, fairness and (social) bias differ. Fairness refers to disparities in the task-specific accuracy of models across social groups. In contrast, bias refers to properties of model generations, i.e. there is no (explicit) relationship with the accuracy or the specifics of a given task.

We study bias in the context of model generations, where we study two such asymmetries. First, we measure bias in *demographic representation*, referring to uneveness in the rates that different demographic groups are mentioned to identify *erasure* and *over-representation*. These measures depend on the occurrence statistics of words signifying a demographic group across model generations. Second, we measure *stereotypical associations*, referring to uneveness in the rates that different groups are associated with stereotyped terms (e.g. occupations) in society. In both cases, by uneveness we mean the extent to which the observed rates diverge from the uniform distribution, i.e. all groups being mentioned or associated with equally, though our metrics allow for alternative references to be considered (see Appendix C.5).

Input: Caspar Weinberger's father, Herman, was the younger
Model generation for input: son of an immigrant grocer.
Perspective API on model generation: Toxic

Input: Karen Kopins Shaw is an American actress
Model generation for input: who has appeared on film including A Nightmare on Elm Street 2: Freddy's Revenge.
Perspective API on model generation: Non-Toxic

Input: His wife was pregnant at the time of the Queen's death
Model generation for input: , and the couple had a son, Edward.
Perspective API on model generation: Non-Toxic

Toxicity = Toxic / (Toxic + Non-Toxic) = 1/3

Figure 21: **Toxicity Metrics.** A demonstration of how we measure toxicity of language model predictions.

These measures dependence on the cooccurence statistics of demographic words with these stereotyped terms across model generations (see Figure 20). We note that such count-based measures can be brittle in several ways in general. Of specific importance for social bias, we emphasize the differential linguistic *marking* of social groups (e.g. "female nurse" vs "male nurse" may be differentially marked due to sociocultural presuppositions and stereotypes) (Rauh et al., 2022).

We report measures of binary gender bias and racial bias, though we encourage future work to explore measures of other social biases, especially given we release all model generations. Since these metrics are measured over model-generated text, we report these bias-related metrics for all core scenarios involving text generation. In Appendix C.5, we provide the formal definitions of our metrics inspired by Bordia & Bowman (2019), as well as our word lists drawing on prior work (Bolukbasi et al., 2016; Garg et al., 2018; Bommasani et al., 2020) under the recommendations of Antoniak & Mimno (2021).

4.8 Toxicity

Whereas the biases we study reflect *distributional* properties of text, we take toxicity to be an *instance-level* property of text. Models have been shown to generate toxic text when prompted (Gehman et al., 2020), even when this text itself is not toxic (Gehman et al., 2020; Dhamala et al., 2021), and including hateful text directed towards specific groups (e.g. Muslims; Abid et al., 2021). Toxicity itself is a complex construct; here we use the term as an umbrella for related concepts like hate speech, violent speech, and abusive language (see Talat et al., 2017).[45] The notion of toxicity is better addressed with greater context (Pavlopoulos et al., 2020) and with clarity on who is determining toxicity (Sap et al., 2019a; Gordon et al., 2022), which we lack in our broad-coverage evaluation. In line with the recommendation of Rauh et al. (2022), we recognize that this work fails to articulate a more precise (operationalized) definition of toxicity, in part because toxicity measurement is conducted uniformly across disparate scenarios to ensure scalability (cf. Selbst et al., 2019). However, in line with the recommendation of Diaz et al. (2022), we believe this work makes important progress in grounding toxicity evaluations in the context of use cases, though we do believe there is ample room for improvement, especially for evaluations grounded in concrete deployment settings.

To operationalize toxicity measurement, we use the Perspective API (Lees et al., 2022)[46] to detect toxic content in model generations. See Figure 21 for an example; further details are deferred to Appendix C.6.

[45] We will later cover how we operationalize toxicity measurement, which is why we do not endeavor to provide a more crisp characterization as our characterization of the construct is only as good as our operationalization in this context.

[46] `https://perspectiveapi.com/`

The Perspective API is widely used in the toxicity literature with extensive analyses (Hede et al., 2021; Lees et al., 2022), including direct critiques (Rauh et al., 2022). We chose to use the Perspective API because it has been strenuously analyzed and its limitations are well-acknowledged, preferring it for these reasons to newer state-of-the-art toxicity detectors (e.g. the top models on hate speech detection leaderboards[47]). That is, we prefer to use the toxicity detection system with extensive testing and clear measurement of its limitations as opposed to other (potentially better but largely unproven) toxicity detection methods.

Since these metrics are measured over model-generated text, we report these toxicity metrics for all core scenarios involving text generation. Further, since we release all model generations, we hope to directly facilitate future work that explores how the qualitative conclusions drawn regarding toxicity depend on the specified toxicity detection mechanism.

4.9 Efficiency

Efficiency is another important dimension to evaluate language models on, since expensive training and inference costs make models less usable and less accessible to wide swaths of users (Schwartz et al., 2020; Bender et al., 2021; Henderson et al., 2020; Kaack et al., 2021; Strubell et al., 2019; Lacoste et al., 2019; Bommasani et al., 2021, §5.3). For example, users might not want to spend 10× more in terms of time or money on training or inferencing a model if it only improves accuracy on a task by 0.1%. We evaluate the efficiency of language models across both training and inference, examining energy, carbon, and wall-clock efficiency as relevant.

4.9.1 Training efficiency

For each model, we report the **energy cost (in kWh)** of training as well as the **CO_2 emitted (in kilograms)** to train the model as recommended by a growing body of work (Strubell et al., 2019; Lacoste et al., 2019; Anthony et al., 2020; Henderson et al., 2020; Bender et al., 2021; Bommasani et al., 2021, §5.3). Both of these metrics capture the number (for distributed training) and type of accelerators used, while the latter models the environmental impact and also takes into account the types of energy sources used to power model training. We do not report training runtimes for two reasons: (i) they are not widely reported, and (ii) they do not capture the number of accelerators used (i.e. more accelerators could theoretically be used to decrease the training time), which likely varies greatly across model creators.

For energy cost and emissions, we use model creators' reported numbers when available at face value. Where numbers are not reported, we approximate energy cost and emissions with the following calculation *if* details about the hardware used and training duration are available:

$$\begin{aligned} e &= n_{\text{GPU}} W_{\text{GPU}} t_{\text{train}} \text{PUE} \\ e_{\text{CO}_2} &= e c_{\text{region}} \end{aligned}$$

For simplicity, we assume that the accelerators used for training are GPUs, but the above calculations are similar for other accelerators like TPUs. e is the energy used in kWh, n_{GPU} is the number of GPUs used for distributed training (necessary to train the large LMs considered in this work), W_{GPU} is the average power draw of a single GPU in kilowatts over the course of training, and t_{train} is the training time in hours. PUE or Power Usage Effectiveness (Strubell et al., 2019) represents the overhead from datacenter cooling costs as well as other energy costs beyond the GPU's energy draw itself, and is set to 1.1 similar to previous work. e_{CO_2} is then the estimate of carbon emissions; c_{region} is the carbon intensity (kgCO_2 per kWh) of the datacenter in which the model was trained. We use the U.S. national average carbon intensity when the datacenter location is not available. All numbers are approximate due to underlying estimation errors and assumptions, but should be of the right order of magnitude. While others have discussed how the estimation methodology used here might be error-prone (Henderson et al., 2020; Cao et al., 2020), we do not have enough information to use a more fine-grained approach. Strubell et al. (2019), Bender et al. (2021) and Patterson et al. (2021) use similar estimation methodologies but cover a different model set than this work. We outline detailed calculations in Appendix C.7.

[47] `https://paperswithcode.com/task/hate-speech-detection`